knowledge-base-skills — User Manual

Pi Extension for Knowledge-Base-Backed Skills

May 2026

knowledge-base-skills

User Manual

Manage pi skills stored as knowledge base articles — save, list, validate, fix, and install skills from the KB to your pi skill directories.

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1. Introduction

knowledge-base-skills is a pi extension that lets you manage pi skills as knowledge base articles. Instead of writing SKILL.md files directly into `.pi/agent/skills/`, you store them in the knowledge base as structured articles with metadata tags. The extension provides tools to save, list, validate, repair, and install these articles as runnable pi skills.

> **Tip**: This extension follows a save-validate-install workflow. You save a skill to the KB, validate it, fix any issues, then explicitly install it to a project or user skill directory.

Key Features

- **Save** skills as linked KB articles (skill-source + documentation)
- **List** all skill articles with validation status
- **Install** skills from KB to pi's skill directories
- **Fix** broken skill articles (missing tags, damaged frontmatter)

Prerequisites

- pi coding-agent runtime installed
- Knowledge base available at `~/.pi/knowledge-base/` (pi core feature)
- The knowledge-base-skills extension installed via `extension_creator`

2. Quick Start Guide

This section walks through creating, saving, and installing a new skill end-to-end.

Step 1: Write a SKILL.md

Create a skill file with embedded frontmatter containing `name` and `description`:

```
---
name: code-reviewer
description: Reviews code for common issues and suggests improvements.
---

You are a code review expert. When asked to review code, check for:

1. Security vulnerabilities (XSS, injection, etc.)
2. Performance bottlenecks
3. Code style violations
4. Missing error handling
5. Architectural issues

Provide clear, actionable feedback with code examples.
```

Step 2: Save to the Knowledge Base

> **Tip**: Use `scope: local` for project-specific skills (saved to `./knowledge-base/`) or `scope: global` for user-wide skills (saved to `~/.pi/knowledge-base/`).

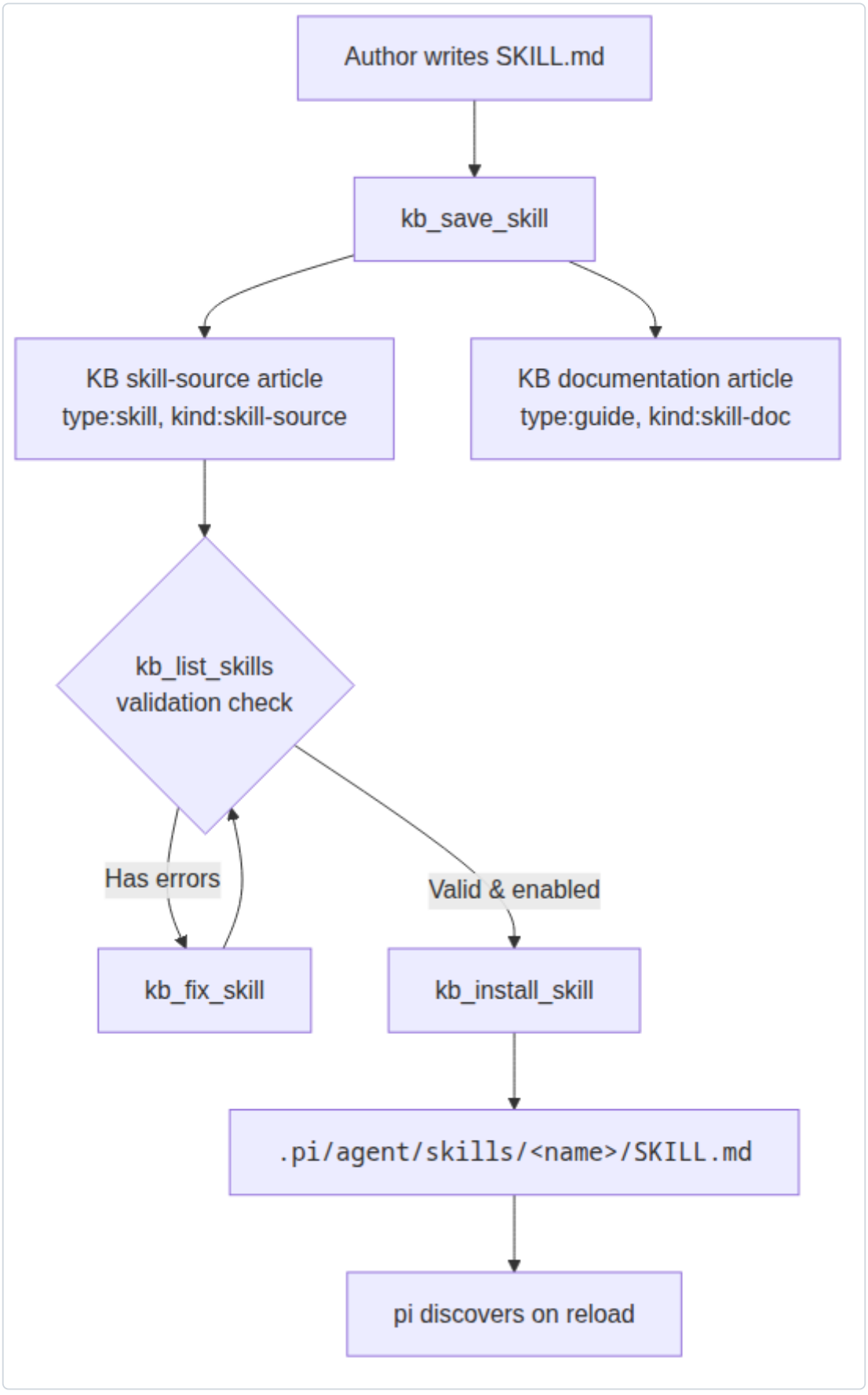
```
kb_save_skill
  skillName: "code-reviewer"
  skillContent: "---\nname: code-reviewer\ndescription: Reviews code for common issues.
\n---\n\nYou are a code review expert..."
  scope: local
  tags: "domain:code-review,tool:read"
```

Result: Two articles are created: - **Skill-source article**: `code-reviewer-skill-source` — contains the raw SKILL.md - **Documentation article**: `code-reviewer-skill` — human-readable reference

Step 3: Verify the Skill

```
kb_list_skills
  scope: local
  verbose: true
```

This lists all skill articles in the local KB with their status. Your new skill should appear as enabled and qualified.



Skill lifecycle diagram

Fig 1: Complete skill lifecycle from creation to installation

Step 4: Install to pi

```
kb_install_skill
  articleSlug: "code-reviewer-skill-source"
  scope: local
```

This writes `SKILL.md` and `SOURCE.json` to `.pi/agent/skills/code-reviewer/`.

Step 5: Reload pi

Run `/reload` in pi to pick up the new skill.

Step 6: Fix Issues (if needed)

If `kb_list_skills` shows validation errors for your skill, fix them before installing:

```
kb_fix_skill
  articleSlug: "code-reviewer-skill-source"
  fixTags: true
  fixFrontmatter: true
  enable: true
```

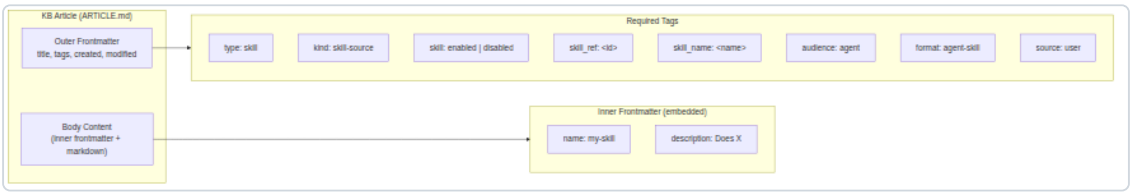
This adds any missing required tags, repairs the inner frontmatter, and ensures the skill is enabled. After fixing, re-run `kb_install_skill` to re-install.

3. Concepts & Architecture

Two-Article Model

Each skill is represented by two linked KB articles sharing a `skill_ref` tag:

Article	Tags	Purpose
Skill-source article	`type:skill`, `kind:skill-source`, `skill:enabled`	Machine-oriented — contains the raw SKILL.md body with embedded frontmatter
Documentation article	`type:guide`, `kind:skill-doc`	Human-oriented — explains usage, examples, links to source



Article data model

Fig 2: Structure of a KB article with outer frontmatter, tags, and inner frontmatter

Inner vs Outer Frontmatter

Skills use two layers of frontmatter:

1. **Outer frontmatter** (article level): `title`, `tags`, `created`, `modified` — parsed by the knowledge base system
2. **Inner frontmatter** (embedded in body): `name`, `description` — parsed by the skill loader

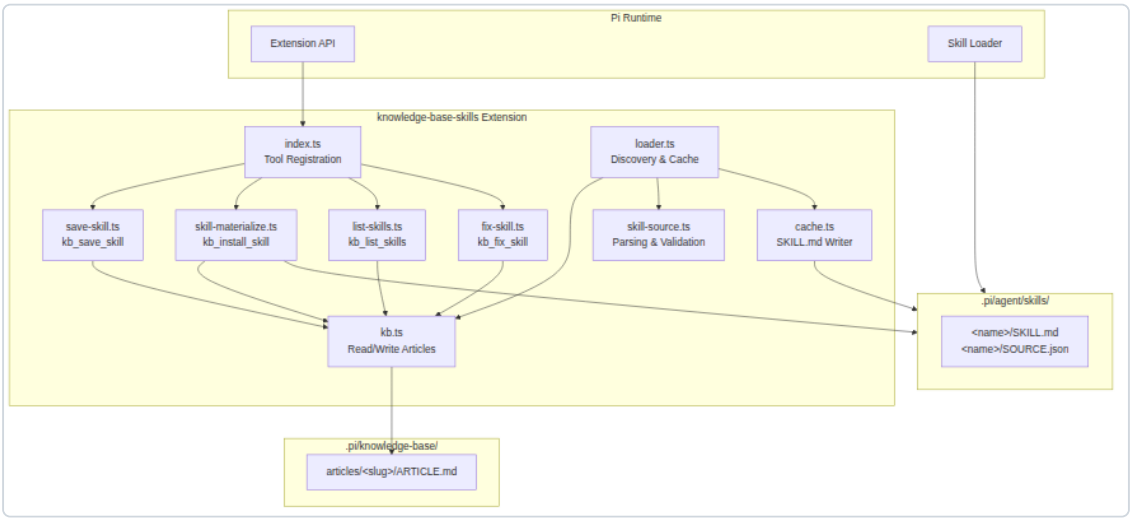
The loader prefers inner frontmatter. If absent, it falls back to the outer article frontmatter fields.

Module Architecture

The extension consists of the following modules coordinated by the entry point:

Module	File	Responsibility
index.ts	Entry	Registers all 4 tools
kb.ts	KB I/O	Read, write, list articles; slug generation; path resolution
save-skill.ts	Save	`kb_save_skill` — creates two linked articles
skill-materialize.ts	Install	`kb_install_skill` — writes SKILL.md + SOURCE.json
list-skills.ts	List	`kb_list_skills` — scans, validates, formats
fix-skill.ts	Fix	`kb_fix_skill` — repairs tags and frontmatter

Module	File	Responsibility
skill-source.ts	Parser	Validates article tags and parses metadata
cache.ts	Cache	Writes SKILL.md files with proper frontmatter
loader.ts	Discovery	Scans KB articles, parses skill sources, writes cache



System architecture

Fig 3: Architecture of the knowledge-base-skills extension, showing module relationships

Tag-Based Qualification

An article qualifies as a skill source only if it has all required tags:

```
type:skill        + kind:skill-source  → is a skill definition
skill:enabled                          → qualifies for installation
skill_ref:<id>    + skill_name:<name>  → provides runtime metadata
audience:agent    + format:agent-skill → machine-oriented content
source:user                            → origin tracking
```

4. Tools Reference

4.1 kb_save_skill — Save a Skill into the KB

Creates a skill-source article and a companion documentation article in the knowledge base.

Parameters

Parameter	Type	Default	Description
`skillName`	string	—	Lowercase kebab-case name (e.g. `code-reviewer`)
`skillContent`	string	—	Full SKILL.md with embedded `name` and `description` frontmatter
`docTitle`	string?	—	Human-friendly title for the documentation article
`docContent`	string?	—	Custom markdown for the documentation article
`scope`	`"local"` / `"global"`	`"local"`	Which knowledge base to save into
`tags`	string?	—	Extra comma-separated `key:value` tags
`enabled`	boolean	`true`	Whether the skill is enabled for loading

Example

```
kb_save_skill
  skillName: "my-analyzer"
  skillContent: "---\nname: my-analyzer\ndescription: Analyzes code structure\n---
\n\nYou analyze code..."
  scope: local
  tags: "domain:code-analysis,tool:read"
  enabled: true
```

Output

```
Saved skill: my-analyzer
  Skill-source article: my-analyzer-skill-source
  Documentation article: my-analyzer-skill
```

4.2 kb_list_skills — List Skill-Source Articles

Scans the knowledge base for skill-source articles and displays their status, description, and any validation issues.

Parameters

Parameter	Type	Default	Description
`scope`	`"local"` / `"global"` / `"all"`	`"all"`	Which KB scope(s) to scan
`status`	`"enabled"` / `"disabled"` / `"all"`	`"all"`	Filter by enabled/disabled
`verbose`	boolean	`false`	Show detailed validation issues per skill

Example

```
kb_list_skills scope: global status: enabled verbose: true
```

Output (verbose)

```
Found 2 skill-source article(s):

     [✓] code-reviewer (local)
       Slug: code-reviewer-skill-source
       Description: Reviews code for common issues and suggests improvements.
       Ref: code-reviewer
       Status: enabled

     [✗] deprecated-analyzer (global)
       Slug: deprecated-analyzer-skill-source
       Description: An old analyzer.
       Ref: deprecated-analyzer
       Status: disabled
          [tag:audience] Missing required tag "audience" with value "agent"
          [name] No skill name found

Summary: 1 qualified, 1 enabled, 1 with errors
```

> **Tip**: Use the non-verbose output for a quick overview, then enable verbose mode to see exactly what needs fixing.

4.3 kb_install_skill — Install a Skill to pi

Materializes a KB skill-source article into a pi skill directory as `SKILL.md` + `SOURCE.json`.

Parameters

Parameter	Type	Default	Description
`articleSlug`	string	—	Slug of the skill-source article in the KB
`scope`	`"local"` / `"global"`	`"local"`	Install target directory

Install Targets

Scope	Directory
`local` (default)	`<cwd>/.pi/agent/skills/<name>/`
`global`	`~/.pi/agent/skills/<name>/`

Example

```
kb_install_skill
  articleSlug: "code-reviewer-skill-source"
  scope: global
```

Output

```
Materialized skill: code-reviewer
  Output: /home/user/.pi/agent/skills/code-reviewer/SKILL.md
```

Generated Files

```
~/.pi/agent/skills/code-reviewer/
├── SKILL.md       ← The skill content with proper frontmatter
└── SOURCE.json    ← Metadata linking back to the KB article
```

Example `SOURCE.json`:

```
{
  "skillName": "code-reviewer",
  "skillRef": "code-reviewer",
  "articleSlug": "code-reviewer-skill-source",
  "articlePath": "/home/user/.pi/knowledge-base/articles/code-reviewer-skill-source/
       ARTICLE.md",
  "materializedAt": "2026-05-10T12:00:00.000Z"
}
```

4.4 kb_fix_skill — Validate and Repair a Skill

Analyzes a skill-source article, identifies missing required tags and frontmatter issues, and repairs them automatically.

Parameters

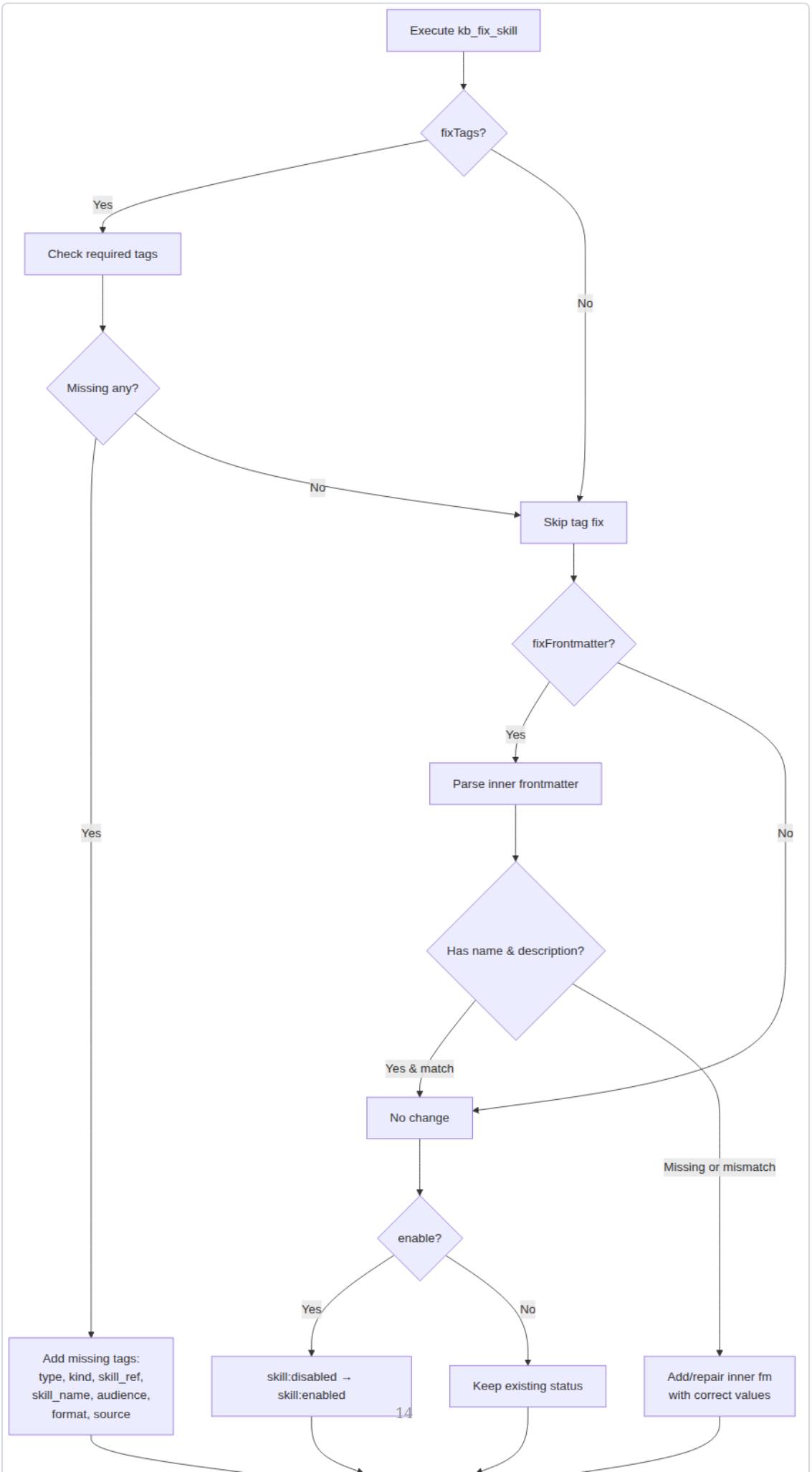
Parameter	Type	Default	Description
`articleSlug`	string	—	Slug of the skill-source article to fix
`fixTags`	boolean	`true`	Add missing required tags
`fixFrontmatter`	boolean	`true`	Add or repair inner embedded frontmatter
`enable`	boolean	—	Change `skill:disabled` → `skill:enabled`
`name`	string?	—	Override skill name (tag + frontmatter)
`description`	string?	—	Override description (frontmatter only)
`source`	string?	`"user"`	Override source tag value

What Gets Fixed

When `fixTags: true` : - Adds `type:skill` if missing - Adds `kind:skill-source` if missing - Adds `skill_ref:<slug>` if missing (uses article slug) - Adds `skill_name:<name>` if missing (uses provided name or slug) - Adds `audience:agent` if missing - Adds `format:agent-skill` if missing - Adds `source:user` if missing (or custom value) - If no `skill` tag exists, adds `skill:disabled` (use `enable:true` to override)

When `fixFrontmatter: true` : - If no inner frontmatter block exists, adds one with `name` and `description` - If inner frontmatter exists but `name` is missing or mismatched, corrects it - If inner frontmatter exists but `description` is missing or mismatched, corrects it - Handles malformed inner frontmatter (e.g., missing closing `—`)

When `enable: true` : - Changes `skill:disabled` → `skill:enabled`


Execute kb_fix_skill
fixTags?
Yes
No
Check required tags
Missing any?
Yes
No
Skip tag fix
fixFrontmatter?
Yes
No
Parse inner frontmatter
Has name & description?
Yes & match
Missing or mismatch
No change
enable?
Yes
No
Add missing tags: type, kind, skill_ref, skill_name, audience, format, source
skill:disabled → skill:enabled
Keep existing status
Add/repair inner fm with correct values

Validation and fix flow

Fig 4: Decision flow for `kb_fix_skill` — shows how each repair option is applied

Example (fix and enable a broken skill)

```
kb_fix_skill
  articleSlug: "deprecated-analyzer-skill-source"
  fixTags: true
  fixFrontmatter: true
  enable: true
  name: "deprecated-analyzer"
  description: "A repaired code analyzer skill"
```

Example Output

```
Fixed skill article: deprecated-analyzer-skill-source
  File: /home/user/.pi/knowledge-base/articles/deprecated-analyzer-skill-source/
ARTICLE.md
  Actions (4):
    • tag_added: Added tag "audience:agent"
    • tag_added: Added tag "format:agent-skill"
    • tag_enabled: Changed skill:disabled → skill:enabled
    • frontmatter_added: Added missing name and description to inner frontmatter
```

5. Skill Article Format

File Structure

```
~/.pi/knowledge-base/articles/<slug>/
└── ARTICLE.md
```

Outer Frontmatter

The outer frontmatter (parsed by the KB system) uses YAML with `title` and `tags` fields:

```yaml
---
title: my-skill — Skill Source
tags:
  - 'type:skill'
  - 'kind:skill-source'
  - 'skill:enabled'
  - 'skill_ref:my-skill'
  - 'skill_name:my-skill'
  - 'audience:agent'
  - 'format:agent-skill'
  - 'source:user'
  - 'domain:code-analysis'
  - 'tool:read'
created: '2026-05-10T12:00:00.000Z'
modified: '2026-05-10T12:00:00.000Z'
---
```

Inner Frontmatter (Embedded)

The body contains the SKILL.md content with its own frontmatter:

```yaml
---
name: my-skill
description: Analyzes code structure and suggests improvements.
---

You are a code analysis expert. When asked to review code ...
```

The inner frontmatter block starts at the beginning of the article body, delimited by `---`. The loader extracts `name` and `description` from this block when materializing the skill.

Complete Example

```yaml
---
title: code-reviewer — Skill Source
tags:
  - 'type:skill'
```

```
  - 'kind:skill-source'
  - 'skill:enabled'
  - 'skill_ref:code-reviewer'
  - 'skill_name:code-reviewer'
  - 'audience:agent'
  - 'format:agent-skill'
  - 'source:user'
  - 'domain:code-review'
  - 'tool:read'
created: '2026-05-10T12:00:00.000Z'
modified: '2026-05-10T12:00:00.000Z'
---

---
name: code-reviewer
description: Reviews code for common issues and suggests improvements.
---

You are a code review expert. When asked to review code, check for:

1. Security vulnerabilities (XSS, injection, etc.)
2. Performance bottlenecks
3. Code style violations
4. Missing error handling
5. Architectural issues

Provide clear, actionable feedback with code examples.
```

When Installed

After installation, the SKILL.md at `.pi/agent/skills/code-reviewer/SKILL.md` looks like:

```
---
name: code-reviewer
description: Reviews code for common issues and suggests improvements.
---

You are a code review expert. When asked to review code, check for...

1. Security vulnerabilities (XSS, injection, etc.)
2. Performance bottlenecks
3. Code style violations
4. Missing error handling
5. Architectural issues

Provide clear, actionable feedback with code examples.
```

The inner frontmatter becomes the outer frontmatter of the installed skill file.

6. Validation & Troubleshooting

Validation Checklist

When you run `kb_list_skills verbose: true`, every skill article is checked against these criteria:

Hard Requirements

Check	Severity	Description
`type:skill` tag present	error	Identifies as a skill definition
`kind:skill-source` tag present	error	Distinguishes from doc articles
`skill:enabled` or `skill:disabled` tag	error	Opt-in flag
`skill_ref` tag non-empty	error	Links source to documentation
`skill_name` tag non-empty	error	Runtime name for the skill
`skill_name` matches `/^[a-z0-9]+(-[a-z0-9]+)*$/`	error	Valid kebab-case, max 64 chars
`audience:agent` tag present	error	Machine-oriented content
`format:agent-skill` tag present	error	Content format marker
`source` tag present	error	Origin tracking
Inner `name` matches `skill_name`	error	Consistency check
Inner or outer `description` non-empty	error	Required metadata
Body content non-empty	error	Skill needs actual content

Warnings

Check	Severity	Description
Using outer frontmatter instead of inner	warning	Inner frontmatter is preferred
Tag value mismatch (e.g., `type:guide`	warning	May cause discovery issues

Check	Severity	Description
instead of `type:skill`)		

Common Issues

Skill Not Listed

Symptom: `kb_list_skills` shows fewer skills than expected.

Causes and fixes: 1. Article lacks `type:skill` and `kind:skill-source` tags - Fix: `kb_fix_skill fixTags: true` 2. Article is in wrong scope - Use `scope: all` to scan both local and global KB 3. Article uses flat file format not in `articles/<slug>/` directory - Move to folder-based layout if required

Skill Not Installing

Symptom: `kb_install_skill` returns “not a valid skill-source article”.

Causes and fixes: 1. Missing required tags - Run `kb_list_skills verbose: true` to see what’s missing - Fix: `kb_fix_skill fixTags: true` 2. Article slug is wrong - Check slug with `kb_list_skills` 3. Article is disabled - Use `kb_fix_skill enable: true` or install directly with `kb_install_skill` (accepts disabled skills)

Broken Frontmatter

Symptom: Installed SKILL.md has no `name` or `description`.

Causes and fixes: 1. Inner frontmatter block is missing from article body - Fix: `kb_fix_skill fixFrontmatter: true` 2. Inner frontmatter has no closing `---` - Fix: `kb_fix_skill fixFrontmatter: true` (handles malformed blocks) 3. Name/description values are in outer frontmatter only - `kb_fix_skill fixFrontmatter: true` migrates them to inner frontmatter 4. Name doesn’t match skill_name tag - Fix: `kb_fix_skill name: "<correct-name>" fixFrontmatter: true`

“description is required” Error

If pi shows this error after installing: 1. Use `kb_fix_skill` to add missing frontmatter 2. Re-install with `kb_install_skill` 3. Run `/reload` in pi

> The skill article in the KB and the installed SKILL.md are separate files. After fixing the KB article, you must re-install the skill for the changes to take effect.

7. Tag Schema Reference

Skill-Source Article Tags

Tag	Required	Value	Purpose
`type`	Yes	`skill`	Identifies as a skill definition
`kind`	Yes	`skill-source`	Distinguishes from doc articles
`skill`	Yes	`enabled` or `disabled`	Opt-in flag
`skill_ref`	Yes	`<id>`	Links source + doc articles
`skill_name`	Yes	`<kebab-name>`	Runtime name
`audience`	Yes	`agent`	Machine-oriented
`format`	Yes	`agent-skill`	Content format
`source`	Yes	`user`	Origin tracking

Documentation Article Tags

Tag	Required	Value	Purpose
`type`	Yes	`guide`	Identifies as guidance
`kind`	Yes	`skill-doc`	Distinguishes from source
`skill_ref`	Yes	`<id>`	Links to source article
`skill_name`	Yes	`<kebab-name>`	Reference to skill
`audience`	Yes	`human`	Human-readable

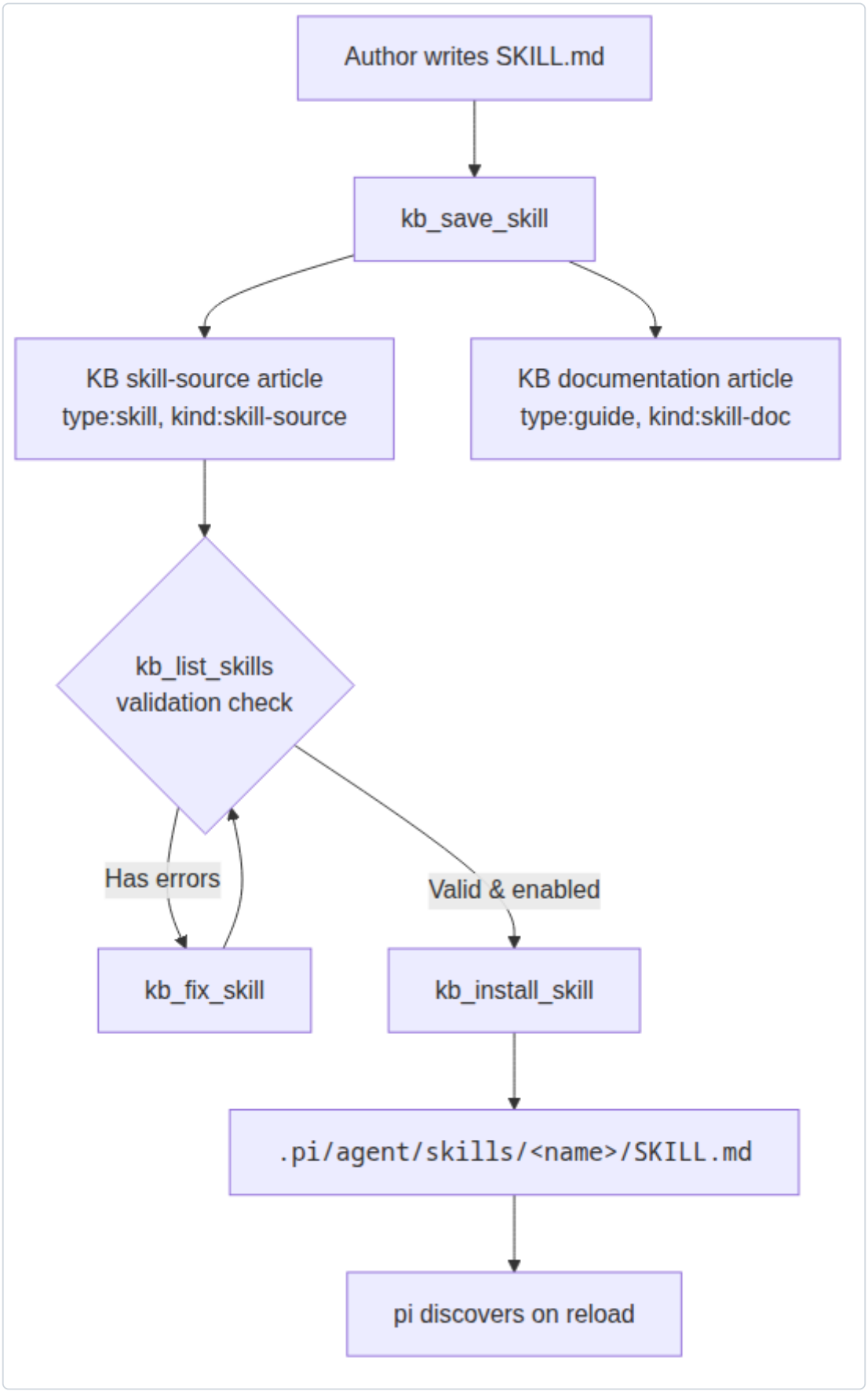
Recommended Tags

Tag	Example	Purpose
`status`	`stable`, `draft`	Lifecycle tracking
`domain`	`code-review`, `testing`	Topic discovery
`tool`	`read`, `bash`, `write`	Related tools
`project`	`knowledge-base-skills`	Project association
`level`	`beginner`, `advanced`	Expertise level

8. Workflow Diagrams

Skill Lifecycle

The diagram below shows the complete path from authoring a SKILL.md to running it in pi:



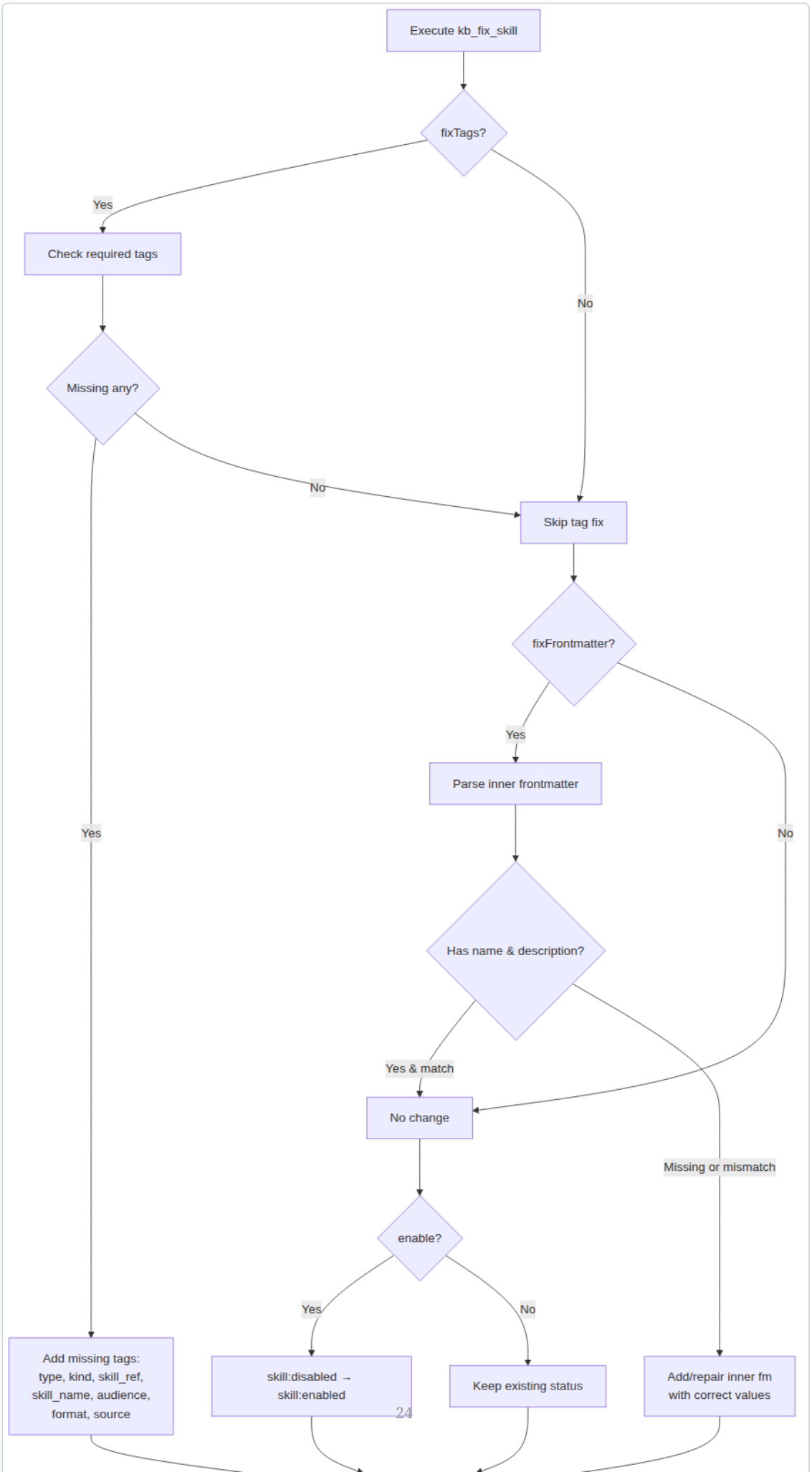
Skill lifecycle workflow

Fig 5: End-to-end skill lifecycle: author → save → validate → fix → install → reload

1. **Author**: Write a SKILL.md with `name` and `description` in embedded frontmatter
2. **Save**: Use `kb_save_skill` to create linked articles in the KB
3. **Validate**: Run `kb_list_skills` to check for issues
4. **Fix** (if needed): Use `kb_fix_skill` to repair missing tags or frontmatter
5. **Install**: Use `kb_install_skill` to materialize into pi's skill directory
6. **Reload**: `/reload` in pi for the skill to take effect

Validation Decision Flow

When you run `kb_fix_skill`, the extension follows a systematic repair process:

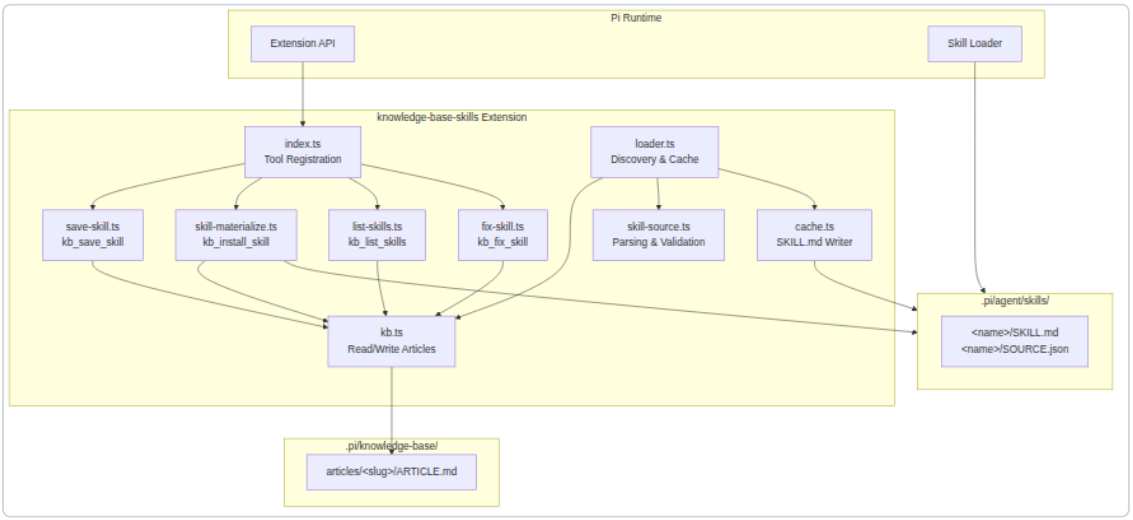

Execute kb_fix_skill
fixTags?
Yes
No
Check required tags
Missing any?
Yes
No
Skip tag fix
fixFrontmatter?
Yes
No
Parse inner frontmatter
Has name & description?
Yes & match
Missing or mismatch
No change
enable?
Yes
No
Add missing tags: type, kind, skill_ref, skill_name, audience, format, source
skill:disabled → skill:enabled
Keep existing status
Add/repair inner fm with correct values

Validation flow diagram

Fig 6: Decision tree for `kb_fix_skill` — tags first, then frontmatter, then enablement

System Architecture

The extension's internal module structure:



System architecture diagram

Fig 7: Module relationships within the extension and with external systems

9. FAQ

Q: What happens if I save a skill with `enabled: false`?

A: The skill is saved to the KB but not flagged for installation. Use `kb_fix_skill enable: true` to activate it, then install with `kb_install_skill`.

Q: Can I store skills in both local and global KB?

A: Yes. Use `scope: local` to save to your project's `./knowledge-base/` directory. Use `scope: global` to save to `~/.pi/knowledge-base/`. The `kb_list_skills` tool can scan either or both.

Q: What's the difference between inner and outer frontmatter?

A: The outer frontmatter is at the KB article level and contains system metadata like `title`, `tags`, `created`, `modified`. The inner frontmatter is embedded in the article body and contains the skill's `name` and `description`. The loader prefers inner frontmatter for skill metadata.

Q: How do I update an existing skill?

A: Edit the ARTICLE.md in the KB directly, then run `kb_install_skill` again to re-materialize. The skill file in `.pi/agent/skills/` is regenerated with the updated content.

Q: Why does `kb_fix_skill` add `skill:disabled` by default?

A: Safety — the tool never enables a skill without explicit consent. Use `enable: true` to flip it to `skill:enabled`.

Q: How can I preview issues without making changes?

A: Run `kb_list_skills verbose: true` to see all validation issues without modifying anything. Then decide which `kb_fix_skill` options to apply.

Q: What if my KB is at a custom path?

A: Set the `KB_SKILLS_KB_PATH` environment variable to override the default `~/.pi/knowledge-base/`. All tools respect this variable.

Q: How do I uninstall a skill?

A: Remove the skill directory from `.pi/agent/skills/<name>/` and run `/reload` in pi. Optionally, delete the KB articles or mark the skill as `skill:disabled`.